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A SML PROJECT REPORT

on

"Statistical Analysis and Machine Learning-Based Prediction of Air Quality Across Indian Regions"

Submitted by

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CERTIFICATE

Certified that the project work entitled **"Statistical Analysis and Machine Learning-Based Prediction of Air Quality Across Indian Regions"** has been carried out by **Prathiksha V (1VI23CD035)**, a bona fide student of Vemana Institute of Technology, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering (Data Science)** of Visvesvaraya Technological University, Belagavi, during the academic year **2026–27**. It is certified that all corrections and suggestions indicated during the internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the said degree.

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ABSTRACT

This project applies statistical analysis and machine-learning methods to 235,785 air-quality observations covering 32 Indian states or regions and 291 areas from 1 April 2022 to 30 April 2025. The notebook examines the Air Quality Index (AQI) distribution, geographic and seasonal patterns, prominent-pollutant labels, and the predictive value of location, calendar, and monitoring-station fields. Its workflow includes data checks and preprocessing, descriptive statistics, correlation, confidence intervals, one-way ANOVA, Tukey post-hoc comparisons, AQI regression, and air-quality-status classification.

The saved outputs report a mean AQI of 111.127 (95% confidence interval: 110.839–111.416) and a median of 92. Seasonal and state/region differences are significant in the reported ANOVAs ($p < 0.001$), with eta-squared values of 0.161 and 0.180. Random Forest Regression gives the best reported regression results (MAE 41.057, RMSE 55.709, $R^2 = 0.241$). Random Forest classification reaches 41.2% accuracy and 0.429 weighted F1, but its macro F1 is 0.294, reflecting weak performance for rare status classes. Month is the most influential Random Forest regression feature in the saved ranking.

All 11 saved notebook plot outputs are included in this report. The corresponding 11 images embedded in the supplied project PDF were checked against the notebook outputs and match. The dataset file aqi.csv is referenced by the notebook but is not present in the workspace, so the notebook's saved calculations could not be rerun. The report therefore distinguishes recorded outputs from independent reproduction. The dataset also lacks numerical pollutant concentration and meteorological measurements; pollutant labels are descriptive, not measured concentrations.

Keywords: Air Quality Index, India, statistical analysis, ANOVA, regression, classification, explainable machine learning.

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1 Project Introduction and Scope

1.1 Project context

Air quality varies over time and across regions. Statistical summaries can describe these patterns, and predictive models can assess whether location, calendar, and monitoring coverage carry useful information about AQI. This report follows the analysis implemented in the supplied AQI notebook, rather than introducing a new dataset or claiming a clinical or operational forecasting application [1]. India's National Air Quality Index communicates ambient air-quality categories and associated health impacts; it is distinct from pollutant-concentration measurements [2].

The observations illustrate how statistical learning can summarize a large environmental dataset: descriptive measures condense distributions, inferential tests compare groups, and predictive models evaluate whether recorded inputs carry information about AQI. These stages answer distinct questions. A difference in group averages does not by itself imply accurate individual predictions, and predictive performance does not establish why a regional or seasonal pattern occurs.

1.2 Problem statement

The project asks how observed AQI varies by region and time, which prominent-pollutant labels occur most often, and whether numerical AQI or air-quality status can be predicted from the non-leaking information available in the dataset. The notebook uses state, season, month, day of week, and number of monitoring stations as model inputs. It does not contain numerical pollutant concentrations or meteorological measurements.

The project therefore treats AQI as a numeric response and status as a categorical response, while treating prominent-pollutant entries as descriptive labels. The dataset does not include measured pollutant concentrations needed to compare directly with concentration-based air-quality guidelines [3]. It is limited to associations supported by the recorded columns and cannot isolate the contributions of weather, emissions, population, or other unrecorded factors.

1.3 Objectives

The notebook's objectives are to:

1. inspect the structure and quality of the Indian AQI dataset;

2. summarize the AQI distribution and its regional and temporal variation;
3. describe prominent-pollutant label frequencies and associated AQI;
4. test for AQI differences across seasons and states/regions;
5. compare models for numerical AQI prediction and status classification; and
6. interpret model performance and identify data and validation limitations.

1.4 Scope and intended outcome

The analysis is observational and exploratory. The status classifier excludes aqi_value because status categories are defined from AQI bands. The results reflect one chronological holdout period and one dataset version. They are not a causal assessment, regulatory determination, or public-health decision tool.

The intended outcome is a clear account of the notebook workflow and its saved results, including what each method measures and where its evidence ends. The report distinguishes recorded outputs from independently regenerated results because the source CSV was not supplied.

2 Code Walkthrough and Explanation

2.1 Evidence boundary

The uploaded notebook contains 127 cells, including saved outputs for data checks, statistical analyses, model comparisons, and plots. The notebook and its saved outputs are available, but its referenced aqi.csv input is not present in the workspace. Consequently, the calculations cannot be rerun from the source data. The following code excerpts are taken from the uploaded notebook and are included to document its implemented workflow; saved outputs are not independently recomputed here.

2.2 Libraries, data loading, and cleaning

The notebook uses Pandas and NumPy for data preparation, Matplotlib and Seaborn for visualization, SciPy for statistical tests, Statsmodels for Tukey HSD, and scikit-learn for preprocessing and models [5]. The import cell loads the analysis dependencies used across the notebook:

```
import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
from scipy.stats import f_oneway

from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
from sklearn.metrics import (
    mean_absolute_error, mean_squared_error, r2_score,
    accuracy_score, precision_score, recall_score, f1_score,
    classification_report, confusion_matrix
)
```

The data-loading cell reads aqi.csv and displays its initial shape and sample records:

```
FILE_PATH = "aqi.csv"

df = pd.read_csv(FILE_PATH)

print("Dataset loaded successfully.")
print("Shape:", df.shape)
display(df.head())
```

The notebook parses dates in day-first order, checks the parsed date range, and records missingness and category counts:

```
# Convert date safely. The source uses day-month-year formatting.
df["date"] = pd.to_datetime(df["date"], dayfirst=True, errors="coerce")

print("Parsed date range:")
print(df["date"].min().date(), "to", df["date"].max().date())

print("\nMissing values:")
display(df.isnull().sum().to_frame("Missing Values"))

print("\nUnique states:", df["state"].nunique())
print("Unique areas:", df["area"].nunique())
print("Unique AQI statuses:", df["air_quality_status"].nunique())
```

Preprocessing converts the response and monitoring-station count to numeric values, removes incomplete essential records and duplicates, drops the empty note field, and derives calendar and season features. The saved initial and final data shapes are both 235,785 rows; the final table has 13 columns and no recorded missing values or duplicate rows.

```
# Work on a copy
data = df.copy()

# Numeric conversion
data["aqi_value"] = pd.to_numeric(data["aqi_value"], errors="coerce")
data["number_of_monitoring_stations"] = pd.to_numeric(
    data["number_of_monitoring_stations"], errors="coerce"
)

# Remove invalid essential records
data = data.dropna(subset=[
    "date", "state", "area",
    "number_of_monitoring_stations",
```

```
    "aqi_value", "air_quality_status"
]).copy()

# Remove duplicates
before = len(data)
data = data.drop_duplicates().copy()
removed_duplicates = before - len(data)

# Remove completely empty note column
if "note" in data.columns and data["note"].isna().all():
    data = data.drop(columns=["note"])

# Time features
data["year"] = data["date"].dt.year
data["month"] = data["date"].dt.month
data["day_of_week"] = data["date"].dt.dayofweek
data["day_name"] = data["date"].dt.day_name()

def assign_season(month):
    if month in [12, 1, 2]:
        return "Winter"
    elif month in [3, 4, 5]:
        return "Summer"
    elif month in [6, 7, 8, 9]:
        return "Monsoon"
    else:
        return "Post-Monsoon"

data["season"] = data["month"].apply(assign_season)
```

The saved observations span 1 April 2022 through 30 April 2025 and cover 32 states/regions, 291 areas, and six status classes. AQI ranges from 3 to 500. Matching initial and final row counts indicate that the documented cleaning operations did not reduce the recorded number of observations; this is a statement about saved outputs, not an independent audit of the unavailable CSV. Derived year, month, weekday, and season fields make date information usable in group summaries and model pipelines while retaining interpretable calendar categories.

2.3 Descriptive analysis and visual exploration

The notebook calculates central tendency, dispersion, and quartiles, and uses a histogram and boxplot to inspect distribution shape and outliers. It groups AQI by status, state, area, month, season, and prominent-pollutant label. The mean (111.127) exceeds the median (92), consistent with the right-skew visible in the saved histogram. The plot outputs appear as Figures 3.1–3.7.

```
season_order = ["Winter", "Summer", "Monsoon", "Post-Monsoon"]

season_summary = (
    data.groupby("season")["aqi_value"]
    .agg(["count", "mean", "median", "std"])
    .reindex(season_order)
)

display(season_summary)
```

The histogram and boxplot complement one another: the histogram reveals distribution shape and concentration, while the boxplot summarizes the median, interquartile range, and tails. The difference between the mean and median indicates that high-AQI observations pull the mean upward. Consequently, the median and quartiles help describe a typical record, and high values should not be discarded solely because they are distant from the center.

Table 1: Selected descriptive statistics reported by the notebook

Statistic	Saved value
Number of observations	235,785
Mean AQI	111.127
Median AQI	92.000
Mode	46
Standard deviation	71.447
Minimum and maximum	3 and 500
First and third quartile	59 and 142
Interquartile range	83

2.4 Statistical inference and association tests

The notebook estimates a Student-*t* 95% confidence interval for mean AQI and applies one-way ANOVA to compare AQI among seasons and states/regions. Eta-squared is calculated as an effect-size measure, and Tukey HSD follows the seasonal ANOVA. Effect sizes complement significance tests by expressing the magnitude of observed differences [6]. The reported signif-

icance threshold is $\alpha = 0.05$. Pearson correlations are calculated only for numerical columns available in the file; no pollutant concentration fields are constructed.

The confidence interval quantifies uncertainty around the estimated overall mean under the notebook calculation; its narrow width reflects the large record count, not a guarantee that every location or monitoring condition is represented equally. ANOVA tests whether group means are compatible with a common-mean null hypothesis. Eta-squared complements the test statistic by describing the share of observed variation associated with the grouping factor. With a large dataset, p-values should be read alongside effect sizes, group sizes, and descriptive differences.

```
season_groups = [
    data.loc[data["season"] == season, "aqi_value"]
    for season in season_order
]

season_f, season_p = f_oneway(*season_groups)

print(f"F-statistic = {season_f:.3f}")
print(f"p-value = {season_p:.6g}")

if season_p < 0.05:
    print("Decision: Reject H0.")
    print("Conclusion: AQI differs significantly across seasons.")
else:
    print("Decision: Fail to reject H0.")
    print("Conclusion: No statistically significant seasonal difference was detected.")
```

2.5 Encoding, selection, and PCA

For regression and classification, the notebook one-hot encodes categorical state and season fields. Classification pipelines standardize month, day of week, and monitoring-station count. AQI regression uses the same categorical encoding and passes numerical fields through. No PCA or separate filter-based feature-selection results are saved in the AQI notebook.

One-hot encoding represents each category with indicator columns instead of implying an artificial ordering among states or seasons. Standardization places numeric classifier inputs on comparable scales, which is particularly relevant to Logistic Regression. The fitted pipelines keep transformations associated with the estimator so the same operations are applied to training and test records. Monitoring-station count is a coverage indicator, not a direct pollution measurement.

```
categorical_clf = ["state", "season"]
numeric_clf = ["month", "day_of_week", "number_of_monitoring_stations"]

preprocess_clf = ColumnTransformer([
    ("cat", OneHotEncoder(handle_unknown="ignore", drop="first"), categorical_clf),
    ("num", StandardScaler(), numeric_clf)
])
```

2.6 Holdout evaluation and model tuning

The executed regression and classifier code uses a chronological split: observations before 1 January 2024 form the training set, and observations on or after that date form the test set. The saved classification test set contains 115,075 observations. The notebook evaluates a mean baseline, Multiple Linear Regression, Random Forest Regression, Logistic Regression, and Random Forest classification with fixed model settings; it does not document a grid-search or cross-validation tuning run.

There is an inconsistency within the notebook: a narrative paragraph says stratify=y was used for classification, but the executed cell actually splits by date and contains no stratified split. The chronological code and saved outputs are treated as the operative experiment here, and the inconsistency is recorded as a reproducibility caveat rather than silently resolved.

A date-based holdout asks how a model trained on earlier records performs on later ones, preserving temporal order rather than randomly mixing past and future. A single cutoff provides only one view of temporal generalization; performance may vary with the test period, regional coverage, and status-label distribution. Fixed model settings permit comparison of the reported estimators, but do not establish that those settings are optimal.

```
train_reg = reg_df[reg_df["date"] < "2024-01-01"].copy()
test_reg = reg_df[reg_df["date"] >= "2024-01-01"].copy()

features_reg = [
    "state", "season", "month", "day_of_week",
    "number_of_monitoring_stations"
]
target_reg = "aqi_value"
```

```
rf_reg = Pipeline([
    ("preprocess", preprocess_reg),
    ("model", RandomForestRegressor(
        n_estimators=100,
```

```
        random_state=42,
        n_jobs=-1,
        max_depth=20,
        min_samples_leaf=2
    ))
])

rf_reg.fit(X_train_reg, y_train_reg)
pred_rf_reg = rf_reg.predict(X_test_reg)

mae_rf_reg = mean_absolute_error(y_test_reg, pred_rf_reg)
rmse_rf_reg = np.sqrt(mean_squared_error(y_test_reg, pred_rf_reg))
r2_rf_reg = r2_score(y_test_reg, pred_rf_reg)
```

2.7 Ablation, explanations, and error analysis

The notebook does not implement a cumulative feature-group ablation. It does provide Random Forest regression feature importances, regression residual diagnostics, class-wise classification reports, and confusion matrices. These artifacts are discussed in Section 3 without inventing feature-group scores. Feature importance is interpreted as model-specific predictive association, not causation.

Residuals can expose systematic prediction errors hidden by a single average metric, while a confusion matrix shows which status categories are confused. These diagnostics describe different aspects of model behavior and should be read alongside aggregate error and class-wise scores. The notebook does not report a controlled cumulative feature-group ablation, so no incremental benefit is attributed to individual groups.

```
print("Logistic Regression Classification Report")
print(classification_report(y_test_clf, pred_logistic, zero_division=0))

print("\nRandom Forest Classification Report")
print(classification_report(y_test_clf, pred_rf_clf, zero_division=0))
```

3 RESULTS AND DISCUSSION

3.1 Dataset profile and exploratory findings

The saved dataset contains 235,785 observations over three years. AQI ranges from 3 to 500, with mean 111.127, median 92, and interquartile range 83. The class counts show that Satisfactory (88,897) and Moderate (77,537) are most common, while Severe (555) and Very Poor (5,671) are comparatively rare. This imbalance is relevant to classifier performance.

The histogram and boxplot show why AQI should not be reduced to a single center measure: a long upper tail can increase the mean while most records remain closer to the median. The class-count plot separately shows how often each status label appears. A classifier can achieve a reasonable weighted score by doing well on common classes while still missing rare ones. Regression errors also span a broad response range, so equal absolute errors may have different significance at low and high AQI values.

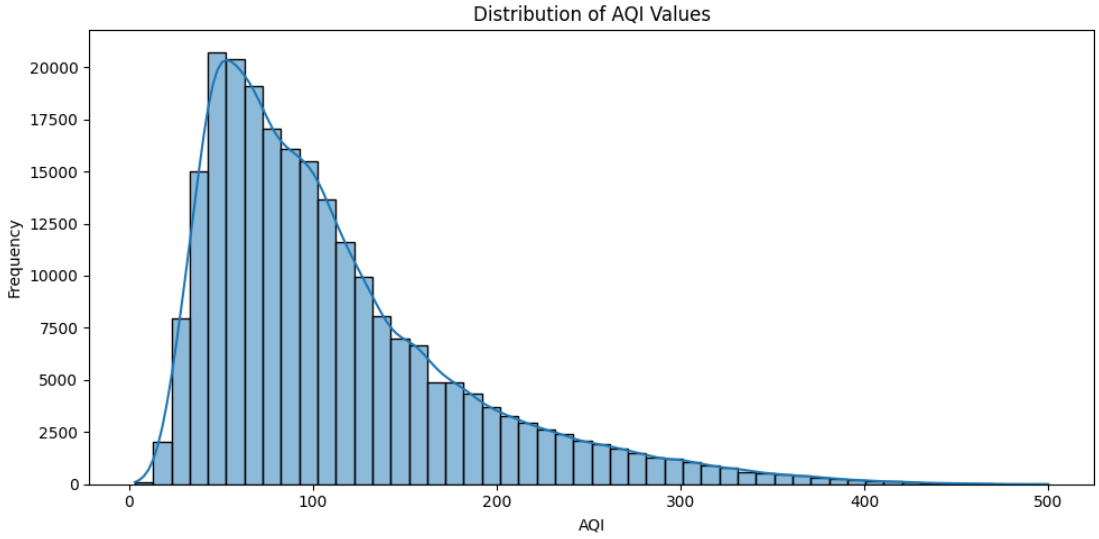


Figure 3.1: Histogram of AQI observations in the saved notebook output.

The state summary shows substantial geographic variation. Delhi has the highest reported state mean AQI (206.417) among the states listed, followed by Jharkhand (164.940) and Himachal Pradesh (160.257). State sample counts vary considerably, so mean comparisons should be interpreted with the group sizes in mind. At the area level, the highest mean among areas with at least 30 records is Byrnihat (239.995); the lowest is Tirunelveli (32.110).

State means based on many observations are generally more stable than those based on few observations, and a ranked chart alone does not show uncertainty. The minimum-record rule reduces the influence of very sparse areas but does not make area coverage uniform or remove all sampling uncertainty. These rankings describe the saved dataset and are not definitive com-

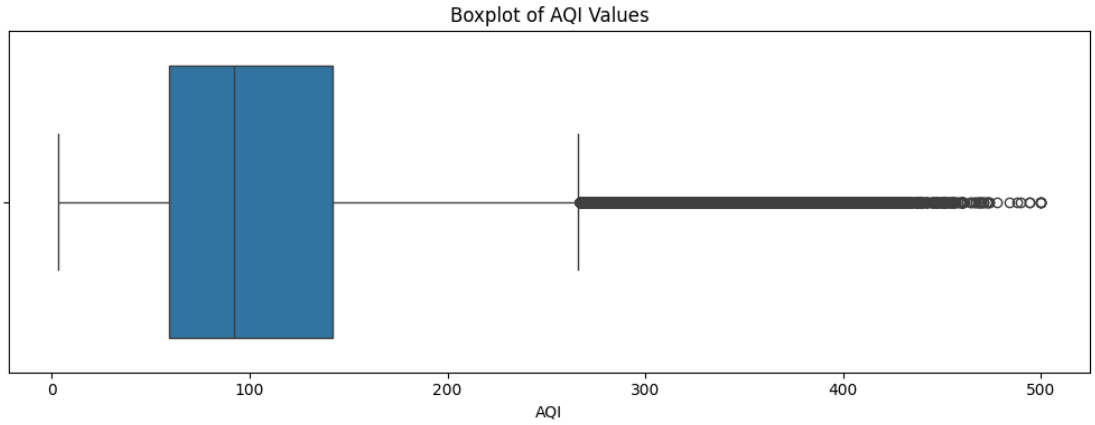


Figure 3.2: Boxplot of AQI values, showing the spread and upper-tail observations.

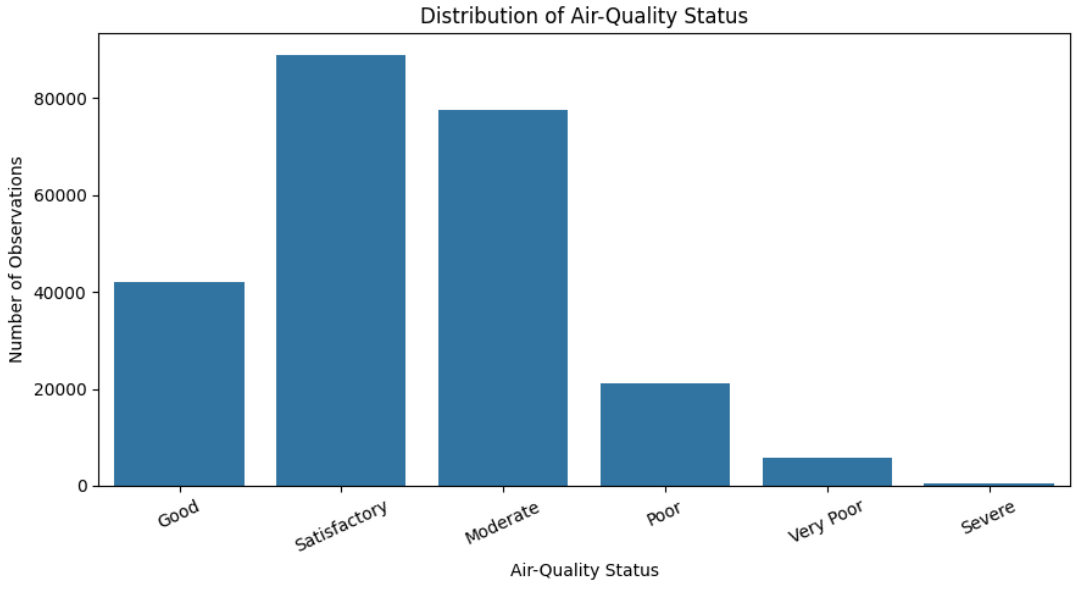


Figure 3.3: Distribution of the six air-quality status classes.

parisons of all local environments.

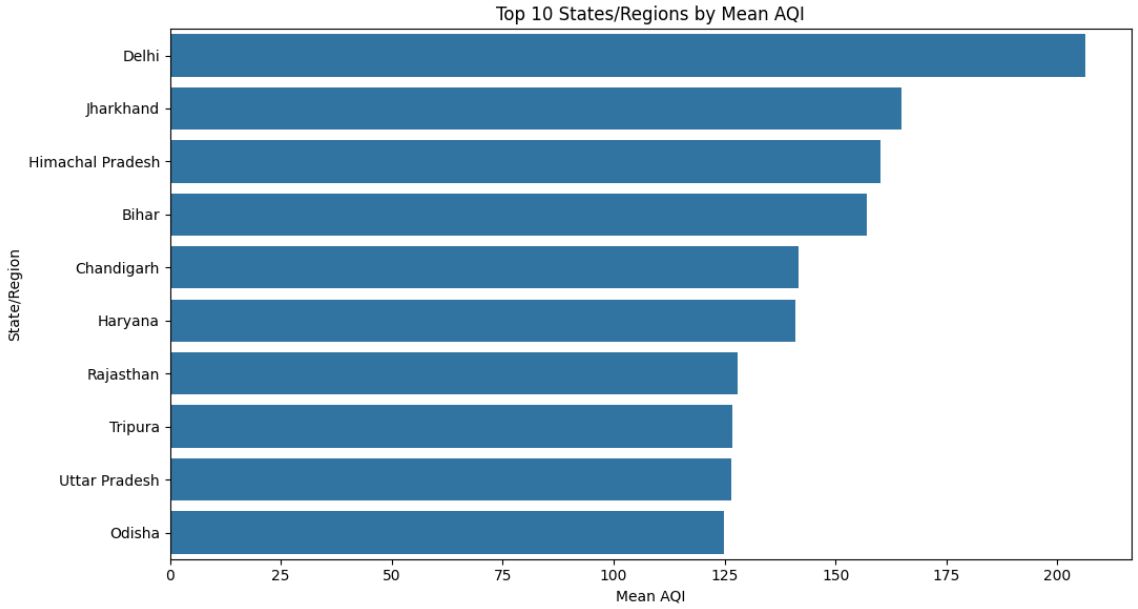


Figure 3.4: Ten states/regions with the highest mean AQI in the saved output.

3.2 Statistical test results

Table 2: Selected statistical results reported by the AQI notebook

Analysis	Result	p-value	Effect size
Mean AQI, 95% CI	111.127 (110.839, 111.416)	–	–
Seasonal ANOVA	$F = 15{,}029.338$; seasonal mean AQI differs	< 0.001	$\eta^2 = 0.161$
State/region ANOVA	$F = 1{,}674.436$; group means differ	< 0.001	$\eta^2 = 0.180$
Tukey HSD	All six seasonal pairs are marked significant in saved output	$p_{adj} < 0.001$	–

The saved seasonal means are highest in Winter (143.128), followed by Post-Monsoon (134.350), Summer (111.387), and Monsoon (72.121). Tukey HSD marks each seasonal pair as significantly different. Given the large record count, effect sizes and descriptive differences are considered alongside small p-values.

Winter exceeds Monsoon by about 71 AQI points in the saved means, indicating that the seasonal contrast is not represented by significance alone. The state/region ANOVA indicates that the observed means are not all equal, but an omnibus test does not identify every pairwise difference. Pairwise conclusions require a suitable post-hoc procedure and attention to multiple

comparisons. The report retains the Tukey result as saved and does not infer unreported state-level comparisons.

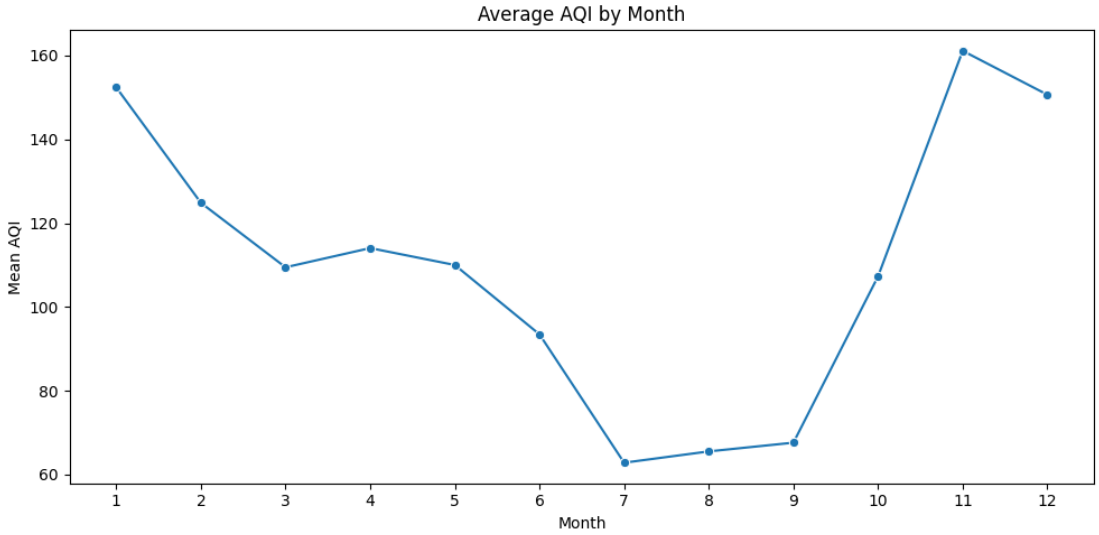


Figure 3.5: Mean AQI by month in the notebook output.

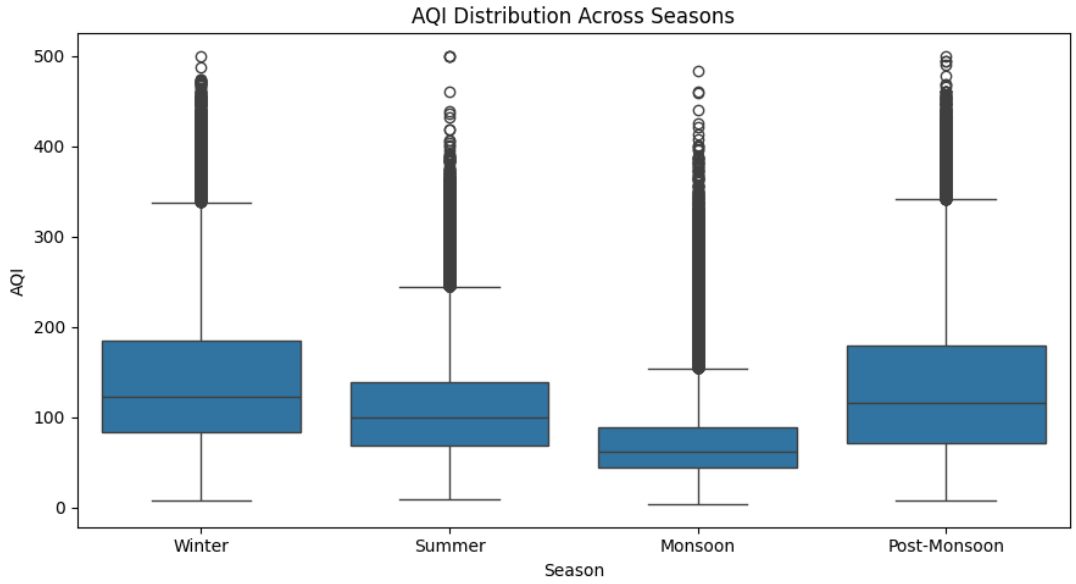


Figure 3.6: AQI distribution across the four seasons.

Prominent-pollutant records can contain more than one label. PM10 is the most frequent label (135,806 records), followed by PM2.5 (78,030). The reported mean AQI is 96.122 for PM10-labeled records and 157.229 for PM2.5-labeled records. These are descriptions of labels and associated AQI, not numerical measures of pollutant concentrations or evidence of pollutant causation.

The label identifies a prominent pollutant category recorded for an observation; it is not a concentration reading and may not represent the full pollutant mixture. A record may contain more

than one label, so counts can overlap and should not be summed as mutually exclusive classes. Group-mean differences may also reflect location, season, or monitoring coverage rather than the label itself.

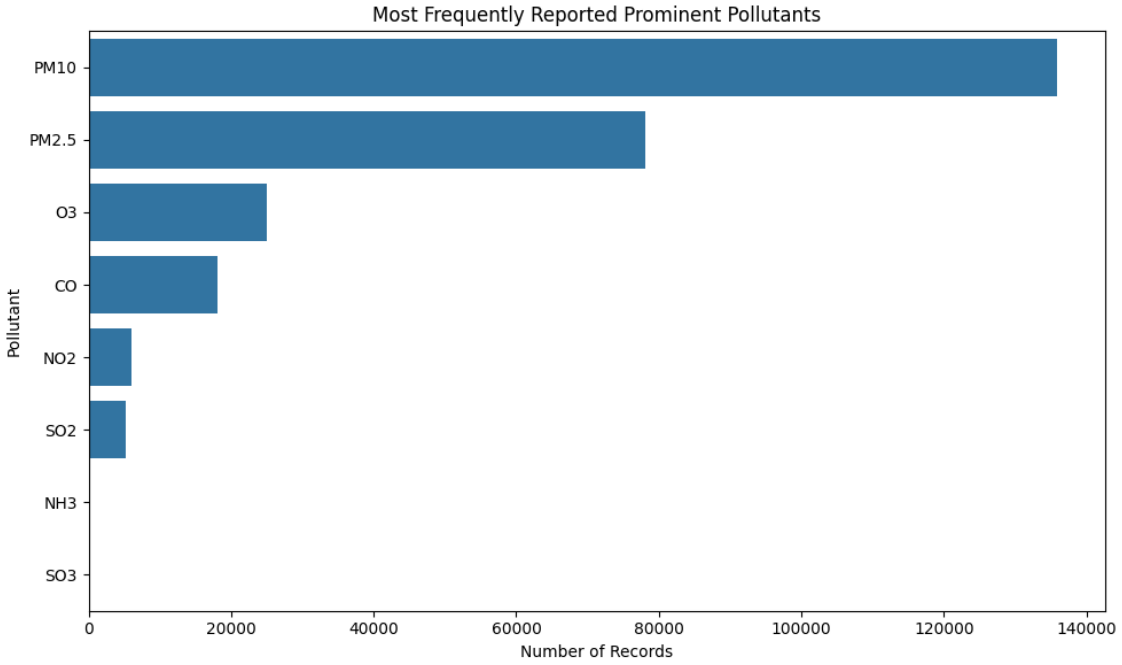


Figure 3.7: Most frequently reported prominent-pollutant labels.

Correlations between AQI and the available numeric variables are weak: approximately 0.077 with monitoring-station count, −0.036 with year, −0.010 with month, and 0.002 with day of week. Correlation is descriptive and does not establish a causal relationship.

The small values indicate little linear association between AQI and these numeric fields under a pairwise measure. They do not rule out nonlinear relationships, interactions, or associations with omitted variables. A numeric month correlation, for example, does not fully represent a repeating annual cycle. The matrix is a diagnostic summary of available columns, not a complete feature-selection verdict.

3.3 Classifier performance

AQI regression and status classification are both reported here. For regression, the mean baseline is outperformed by both fitted models. Random Forest Regression, an ensemble method based on collections of decision trees [4], has the lowest reported MAE and RMSE and the highest R^2, although its $R^2 = 0.241$ means most holdout variation remains unexplained.

MAE summarizes average absolute prediction error in AQI units. RMSE also uses AQI units but penalizes larger misses more heavily because errors are squared before averaging. The

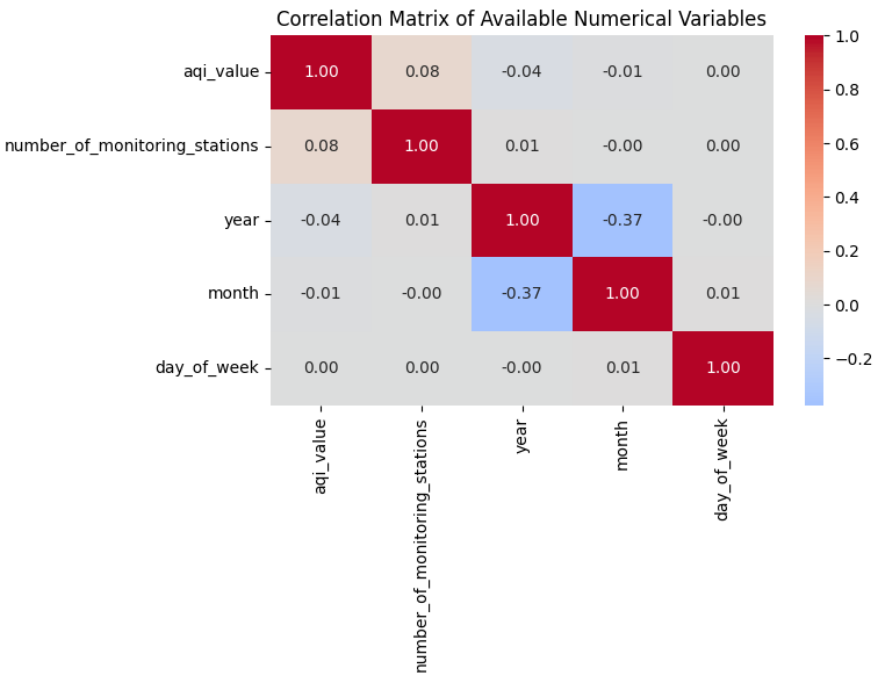


Figure 3.8: Correlation matrix for the numeric variables available in the dataset.

coefficient of determination compares the model with a mean-based reference on the test set; its positive but modest value indicates improvement over that reference while leaving substantial residual variation. Together, the metrics favor the Random Forest among the reported regressors but do not support high-precision individual forecasts.

Table 3: Saved holdout regression and classification metrics

Model and task	Accuracy	MAE	RMSE	R^2 / weighted F1
Mean baseline (regression)	–	50.725	64.301	$R^2 = -0.011$
Multiple Linear Regression	–	43.912	56.124	$R^2 = 0.230$
Random Forest Regression	–	**41.057**	**55.709**	$R^2 = 0.241$
Logistic Regression (status)	0.394	–	–	F1=0.427
Random Forest Classifier (status)	**0.412**	–	–	F1=**0.429**

The regression residual plot shows increasing residual spread as predicted AQI rises, consistent with the notebook's stated concerns about non-constant error variance and missed nonlinear structure.

A residual pattern can indicate that prediction errors depend on the fitted level rather than behaving as a uniform cloud around zero. Here, the widening spread suggests typical error may differ across the AQI range. The diagnostic motivates further modeling and validation, but does not establish which alternative model would perform better; that requires a new, properly held-out comparison.

Random Forest classification has slightly higher accuracy and weighted F1 than Logistic Re-

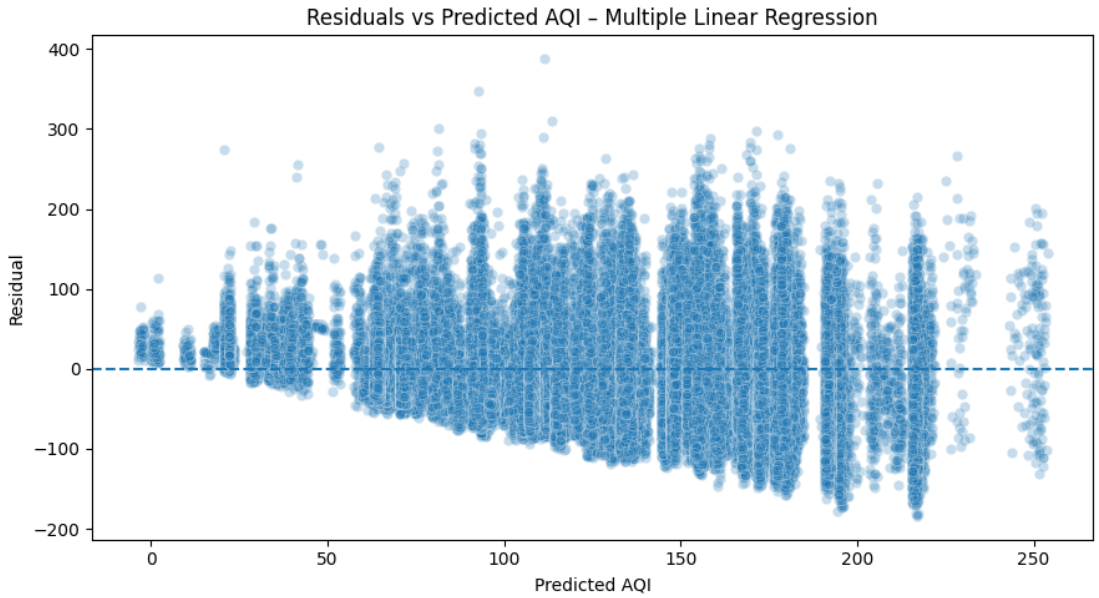


Figure 3.9: Residuals versus predicted AQI for Multiple Linear Regression.

gression. Its macro F1 is 0.294, compared with 0.283 for Logistic Regression, reflecting the influence of class imbalance. The class-wise report shows F1 of 0.01 for the very rare Severe category for both classifiers; weighted scores alone would obscure this limitation.

Accuracy is the share of all predictions that are correct, while weighted F1 aggregates class-wise F1 scores in proportion to class support. Macro F1 gives every class equal weight and is more sensitive to poor performance on rare labels. The report shows the distinction: Moderate has substantial support and a moderate F1, while Severe has only 68 test records and a near-zero F1 despite nonzero recall. Rare-class estimates are uncertain, but this weakness should not be hidden by a weighted average.

Table 4: Random Forest status-classification report on the saved test set

Status	Precision	Recall	F1-score	Support
Good	0.48	0.59	0.53	19,799
Moderate	0.51	0.46	0.48	40,148
Poor	0.18	0.41	0.25	9,260
Satisfactory	0.57	0.29	0.39	44,239
Severe	0.00	0.31	0.01	68
Very Poor	0.07	0.24	0.10	1,561

3.4 Cumulative feature-group experiment and model explanation

No cumulative feature-group ablation is implemented in the AQI notebook. The comparison is therefore not fabricated; Table 5 records the available model explanation instead. Month dom-

inates the Random Forest regression importance ranking, followed by selected state indicators. These importances describe reliance within this fitted model, not causes of AQI.

High importance for month means the fitted forest relied on that feature under its importance calculation; it does not mean changing the calendar month causes a corresponding AQI change. Related predictors can share or redistribute importance, and one-hot encoding creates separate indicators for individual states. The table records which inputs the saved model emphasized, while causal interpretation and the unimplemented feature-group ablation remain outside the notebook evidence.

Table 5: Feature explanation and cumulative feature-group evidence

Random Forest regression feature	Importance
Month	0.400
State: Bihar	0.157
State: Karnataka	0.094
State: Haryana	0.051
Monitoring-station count	0.040
Cumulative group ablation	Not implemented in AQI notebook

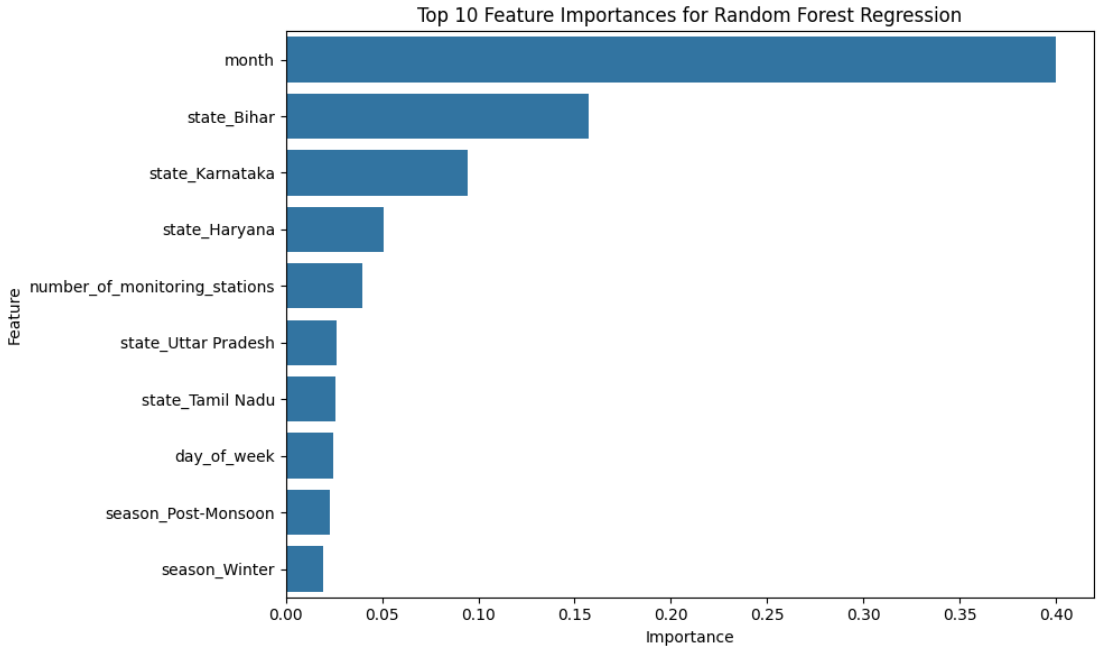


Figure 3.10: Top ten Random Forest regression feature importances saved in the notebook.

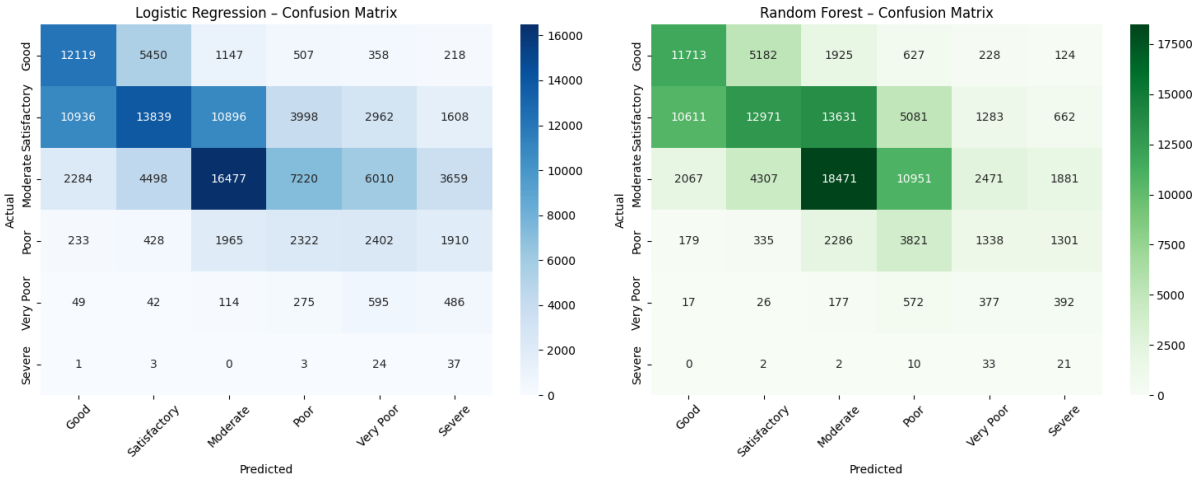


Figure 3.11: Logistic Regression and Random Forest classification confusion matrices.

The diagonal cells represent correct predictions, while off-diagonal cells show where the models confuse one status with another. Both matrices show clearer diagonal patterns for the more common classes and weaker performance for rare categories. This agrees with the class-wise results in Section 3.3: Random Forest F1 is 0.53 for Good and 0.48 for Moderate, but only 0.01 for Severe and 0.10 for Very Poor. Its higher overall accuracy and weighted F1 therefore do not indicate equally reliable performance across all six classes.

4 SOURCE COMPARISON AND REPRODUCIBILITY NOTES

4.1 Evidence limitations

The uploaded AQI notebook is the source for the code and saved outputs. The uploaded AQI project PDF contains the same 11 plot images: each embedded PDF image was checked against the corresponding notebook image output and matched pixel-for-pixel. All 11 notebook output plots are included as Figures 3.1–3.11.

Table 6: Uploaded artifact comparison and evidence used in this report

Artifact	Available content	Role and limitation
AQI notebook	Code cells, saved text outputs, 11 saved plot outputs	Primary source for methods and reported numerical results.
AQI project PDF	Exported analysis and 11 embedded plot images	Cross-check for the notebook figures; its plot images match the notebook outputs.
aqi.csv	The dataset source for the project	Required to rerun preprocessing, regenerate outputs, or independently validate saved metrics.

A notebook narrative sentence states that classification used stratification, while its executed split cells use a date cutoff and do not stratify. The report follows the executed chronological split and flags that mismatch. Other limitations recorded by the notebook are the absence of numerical pollutant concentrations and meteorological variables, the status imbalance, and use of a single temporal holdout. The observed model outputs are not independently regenerated because the dataset file is unavailable.

The image comparison verifies the visual artifacts, not the underlying data computations or every textual metric. Reproducibility would be strengthened by archiving the CSV, recording the software environment and random seeds, and rerunning the notebook end-to-end. The discrepancy between the narrative stratification claim and the executed chronological split should also be corrected in the source notebook.

5 CONCLUSION

The notebook's saved results describe a right-skewed AQI distribution across 235,785 observations, with mean 111.127 and median 92. Seasonal and state/region AQI differences are reported as statistically significant, with substantial effect sizes. The prominent-pollutant analysis describes label frequency rather than pollutant concentration, and the available numerical variables have weak linear correlations with AQI.

Together, these summaries show that AQI varies across the recorded time and location groups, while the right-skewed distribution cautions against relying on the mean alone. The limited feature set prevents the analysis from explaining the environmental mechanisms behind the observed patterns.

Random Forest Regression has the best reported regression scores, but its $R^2 = 0.241$ indicates limited predictive explanation. Random Forest classification slightly exceeds Logistic Regression on accuracy and weighted F1; macro F1 and class-specific results show difficulty with rare Severe and Very Poor categories. These findings are useful as an exploratory statistical-learning exercise, but are constrained by unavailable concentration and weather features, the unprovided dataset file, and an inconsistency in the notebook's split description.

The project demonstrates a progression from cleaning and summarization to statistical tests, model evaluation, and interpretation, while showing why validation design and class-wise diagnostics matter. The results should be treated as a baseline for further study, not as evidence that AQI can be reliably predicted for every region or status category from the current predictors.

6 FUTURE SCOPE

Future work should prioritize the following:

The sequence starts with reproducibility and measurement quality, then improves evaluation before increasing model complexity. This order helps ensure that later performance gains are based on a stable comparison rather than changes in data handling or test composition:

- include the exact version of aqi.csv and rerun the notebook end-to-end;
- add validated pollutant-concentration and meteorological measurements;
- resolve and document the classification split choice, then evaluate multiple forward-in-time test periods;
- report per-class metrics and uncertainty, with particular attention to rare status categories; and
- test model stability across regions and periods before considering any operational use.

7 REFERENCES

References

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